Vel Tech Group of Institutions

Name: Gontu Yaswanth
Email: vtu33434@veltech.edu.in
Roll no: Vtu33434
Phone: null
Branch: VTU
Department: CSE
Batch: 2028
Degree: B.Tech - Computer Science and Engineering

2028_NeoColab_Competitive Coding II_L8

VTU_Week 4_COD

Attempt : 1
Total Mark : 50
Marks Obtained : 50

Section 1 : Coding

1. Problem Statement

Neo is experimenting with text transformations and decides to use a stack to reverse the sequence of words in his sentence. He provides a line of text and expects the program to utilize a stack to reorder the words so that the last word appears first and the first word appears last.

Write a program to ensure this is done correctly using a stack-based approach.

Example

Input: I am Neo

Output: Neo am I

Input Format

The input consists of a single line of input containing a sentence with words separated by spaces.

Output Format

The output prints a single line of output containing the words of the sentence reversed in order, separated by a single space.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: Hello World
Output: World Hello

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    char stack[50][50];
    int top = -1;

    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    char *token = strtok(str, " ");

    while (token != NULL) {
        top++;
        strcpy(stack[top], token);
        token = strtok(NULL, " ");
    }

    for (int i = top; i >= 0; i--) {
        printf("%s", stack[i]);
        if (i != 0)
```

```
        printf(" ");
    }

    return 0;
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

Give a string, remove duplicate letters from the string s such that each letter appears just once using Stack. The lexicographical order of your result must be the least of all feasible outcomes.

Input Format

The input consists of a string.

Output Format

The output displays the string in the lexicographical order with unique characters.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: bcabc

Output: abc

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>

int main() {
    char s[30], stack[30];
    int last[26] = {0}, visited[26] = {0};
    int top = -1;
```

```
    scanf("%s", s);
    int n = strlen(s);

    for (int i = 0; i < n; i++)
        last[s[i] - 'a'] = i;

    for (int i = 0; i < n; i++) {
        char c = s[i];

        if (visited[c - 'a'])
            continue;

        while (top >= 0 && stack[top] > c && last[stack[top] - 'a'] > i) {
            visited[stack[top] - 'a'] = 0;
            top--;
        }

        stack[++top] = c;
        visited[c - 'a'] = 1;
    }

    stack[top + 1] = '\0';
    printf("%s", stack);

    return 0;
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Mira is learning about palindromes and wants to build a program that can check whether a given string is a palindrome using a stack-based approach. A palindrome reads the same backward as forward.

Write a program to ensure that the task is done by pushing characters to a stack and comparing them in reverse order to determine if the string is a palindrome.

Input Format

The input contains a single line containing a string with space-separated characters.

Output Format

The output consists of ,

If the string is a palindrome, print "It's a palindrome!"

Otherwise, print "Not a palindrome"

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: madam
Output: It's a palindrome!

Answer

```
// You are using GCC
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
   char str[100], clean[100], stack[100];
   int top = -1, k = 0;

   fgets(str, sizeof(str), stdin);
   str[strcspn(str, "\n")] = '\0';

   for (int i = 0; str[i]; i++) {
      if (isalpha(str[i])) {
         clean[k++] = tolower(str[i]);
      }
   }
   clean[k] = '\0';

   for (int i = 0; i < k; i++) {
      stack[++top] = clean[i];
   }
```

```
    int isPalindrome = 1;

    for (int i = 0; i < k; i++) {
        if (clean[i] != stack[top--]) {
            isPalindrome = 0;
            break;
        }
    }

    if (isPalindrome)
        printf("It's a palindrome!");
    else
        printf("Not a palindrome");

    return 0;
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Gokul is working on a program to rearrange elements in a queue using an array. Given a queue of integers, he needs to modify it by interleaving its first half with the second half. Write a program to help Gokul achieve this task.

Note: Interleaving involves merging elements from two separate halves in an alternating fashion to create a new sequence.

Example

Input

6

1 2 3 7 9 5

Output

1 7 2 9 3 5

Explanation

The interleave rearranges elements in a queue by alternating between two halves. For instance, given input [1, 2, 3, 7, 9, 5], it splits into [1, 2, 3] and [7, 9, 5], then interleaves them as [1, 7, 2, 9, 3, 5]. It achieves this by splitting the queue into two halves and then merging them interleaved.

Input Format

The first line of the input consists of an integer n, representing the number of elements in the queue.

The second line consists of n space-separated integers, representing the queue elements.

Output Format

The output prints n space-separated integers, representing the rearranged queue after interleaving its first half with the second half.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6
1 2 3 7 9 5

Output: 1 7 2 9 3 5

Answer

```
// You are using GCC
#include <stdio.h>

int main() {
   int n;
   scanf("%d", &n);

   int arr[20];
   for (int i = 0; i < n; i++)
      scanf("%d", &arr[i]);

   int half = n / 2;
```

```
  for (int i = 0; i < half; i++) {
    printf("%d %d ", arr[i], arr[i + half]);
  }

  return 0;
}
```

Status : Correct ***Marks : 10/10***

5. Problem Statement

Sanjay is learning about queues in his data structures class and he wants to write a program that counts the number of distinct elements in the queue. Your task is to guide him in this.

Example

Input:

7

45 33 29 45 45 46 39

Output:

5

Explanation:

The distinct elements in the queue are 45, 33, 29, 46, 39. So, the count is 5.

Input Format

The first line of input consists of an integer M, representing the number of elements in the queue.

The second line consists of M space-separated integers, representing the queue elements.

Output Format

The output prints an integer, representing the number of distinct elements in the queue.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
30 24 43 31 30

Output: 4

Answer

```
// You are using GCC
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int arr[100];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int count = 0;

    for (int i = 0; i < n; i++) {
        int isDuplicate = 0;

        for (int j = 0; j < i; j++) {
            if (arr[i] == arr[j]) {
                isDuplicate = 1;
                break;
            }
        }

        if (!isDuplicate)
            count++;
    }

    printf("%d", count);

    return 0;
```

}

Status : Correct *Marks : 10/10*